

Introduction to Machine Learning

HW1: Exploratory Analysis of a Music Catalog

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Course Website



<https://nthu-cs-hmilab.github.io/I2ML-2026/>

Rules

- Individual assignment. **No plagiarism.**
- All kinds of AI tools are permitted. But you must **submit the AI-use log.**
- **Do not use AI tools to generate the content of the report.**
- Report can be either written in **English or Traditional Chinese.**
- No GPU is required. Google Colab is sufficient enough to complete the task.
- Do not use external datasets, features, or information collected from other sources.
- Submit your work on eeclass. **No late submission will be accepted.**

Deadline

Submission	Deadline
A pre-registration report in the PDF format, up to 2 pages . File name: HW1_pre_report_{student_id}.pdf	9/30 11:59 PM
A final report in the PDF format, up to 10 pages . File name: HW1_final_report_{student_id}.pdf	10/8 11:59 PM
An .ipynb notebook containing all code needed to reproduce the results reported in your submission. File name: HW1_code_{student_id}.ipynb	
The AI-use log document. File name: HW1_ai_use_{student_id}.md	

About AI Use Log

- Submit an AI-use log written in the markdown file:
 - What you used AI for.
 - How you checked the output of AI.
- You should be able to explain any AI-assisted work during the demo session.
- A **5-point deduction** will be applied for **missing submissions or conspicuously low-effort responses** (e.g., merely writing "I used ChatGPT").

Materials

- We provide the starter notebook and the raw dataset.
- Please check out eeclass or course website for more details.

Introduction

You are given a music catalog containing **114,000 track records**. Suppose a team ultimately wants to use this dataset to build machine-learning systems for tasks such as **predicting track popularity**. Before deciding what models to build, the team first needs to understand the dataset itself:

- Is the **dataset reliable** enough to use as training data?
- What **structure** is present in the audio-feature space?
- Which **patterns** appear to reflect meaningful properties of the tracks, and which may instead result from the way the dataset was collected or represented?
- Which target variables can be **predicted** meaningfully from the available audio features, and which cannot?

Your goal in this homework is to answer these questions **with evidence**.

Purpose of the Homework

- **Not to obtain the highest possible accuracy.** You will not compete with your classmate!
- We expect you to:
 - **Explain the reasons** behind your analysis choices
 - **Recognize** when the data support / not support the conclusion.
 - A weak or unsuccessful result may be a correct finding if you **demonstrate** it thoroughly.
- **Feel free to explore any possibilities inside the provided dataset.**



Section 1: The Data

- Dataset of spotify tracks over a range of genres.

- The raw dataset is available from huggingface:

<https://huggingface.co/datasets/maharshipandya/spotify-tracks-dataset>

- Contain 114,000 rows and 20 columns.

Columns	Type	Description
<code>track_id, artists, album_name, track_name</code>	String	Track identifiers and metadata
<code>track_genre</code>	String	The genre category associated with the track
<code>popularity</code>	Integer	A popularity score provided by the streaming platform (0~100)
<code>duration_ms</code>	Integer	Track duration in milliseconds
<code>explicit</code>	Boolean	Indicator for explicit lyrics
<code>danceability, energy, speechiness, acousticness, instrumentalness, liveness, valence</code>	Float	Audio features automatically extracted from the track's audio
<code>loudness</code>	Float	Estimated loudness of the track
<code>tempo</code>	Float	Estimated tempo in BPM
<code>key, mode, time_signature</code>	Integer	Musical attributes stored as integer-coded values

Section 2: Pre-Registration (5%)

- **Submit this report before 9/30 11:59 PM. At most two pages.**
- **There are no formal answers!**
- **Make your expectations** before you analyze the full dataset. **Do not compute summary statistics, create plots, search the full dataset, or fit any models before submitting this section.**
- Inside your report:
 - Two testable expectations about the dataset.
 - One data-quality problem you think might be in the dataset.
 - Predicted held-out error for two regression tasks.
 - One question about the dataset, labels, or data-collection process.

Section 3: Auditing the Dataset (12%)

- Real-world datasets often contain problems that can affect a machine-learning analysis, for example, **missing or invalid values, repeated records, inconsistent entries, or variables whose meaning is easy to misinterpret.**
- Your task in this section is to **identify** the most important **data-quality problems**:
 - **Prioritize at least four data-quality findings.** For each finding, include:
 - The evidence (a table or a figure to show the issue)
 - Why it matters (explain why it affects the pipeline)
 - Your decision (state what you will do about it)
 - Trade-off (explain the information you may loss due to your decision)
 - **Justify your ranking.**
 - **Apply your cleaning decisions** to the pipeline.

Do not miss any one of them, they are the most valuable lessons you will learn from the homework!!!

Section 4: The Geometry of the Feature Space (10%)

- The numeric **audio features** place each track at a point in a multidimensional feature space. In this section, use **principal component analysis (PCA)** to understand the main directions in which the tracks vary, how many of those directions are important, and how feature scaling affects the result.
- Use the cleaned dataset you processed after section 3.

Section 4: The Geometry of the Feature Space (10%)

1. **Effective dimensionality**: Use PCA to estimate how many principal components are needed to represent most of the variation **in the audio features**.
2. **Interpret the leading principle components**: For the first few principal components, explain in plain language what kind of variation each component appears to capture.
3. **Effect of standardization**: Repeat the PCA without standardizing the features first. Compare the result with the standardized PCA and explain why they differ.

Section 5: Structure in the Data (10%)

- Investigate whether tracks with similar audio features naturally form groups, and whether those groups **reflect meaningful musical structure** rather than artifacts of how the dataset was collected.
- Use the cleaned dataset you processed after section 3.

Section 5: Structure in the Data

1. Clustering genres:

- a. Use **hierarchical clustering** to group similar genres together to create one representative audio-feature profile for each genre.
- b. Show how the resulting genre groupings differ in terms of:
 - i. Two different **measures of dissimilarity**.
 - ii. Two different **linkage methods**.

2. Clustering individual tracks:

- a. Cluster with **k-means**, and choose at least **two quantitative criteria** (e.g. inertia curve and the silhouette score) over a range of k.
- b. Describe what each resulting cluster represents in terms of its audio characteristics.

3. Are the clusters meaningful?

- a. Provide evidence that the clusters you found in *Question 2* reflect real differences between tracks, rather than being an artifact of the algorithm, the sampling, or the way genres were assigned.

Section 6: Two Regression Targets (13%)

- You will build regression models for **energy** and **popularity** features.
- Use a **held-out test set** to evaluate predictive performance.
- **Do not include the target itself among the input features.**
- Reminder:
 - Please **watch out for the available features you can use as the input** (as mentioned in the spec).
 - Notice that **energy** feature can be used as input only in building regression model for **popularity** target.
- **Hint: Watch out for data leakage during train-test splitting.**

Section 6: Two Regression Targets (13%)

1. Compare your pre-registration predictions with the actual results:
 - a. Fit a **linear regression model** for each target and report its **held-out RMSE**.
 - b. Report the RMSE of a **simple baseline model** that ignores the input features and always predicts the mean target value from the training set.
2. Compare the two regression tasks and explain why one target is easier or harder to predict from the available features.
3. If one target is difficult to predict from the audio features, look for one other variable in the dataset that may contain useful information about that target.
4. Study the effect of regularization.
 - a. Fit a **regularized linear regression model**.
 - b. Vary the **regularization strength**.

Grading Policy

- **Report 50% + Demo 50%**
- To prevent grader bias, each section is assigned to a dedicated TA to grade.

Tier	Description	Percentage
4	Provides evidence, explanation, and decisions , along with a discussion of limitations or alternative interpretations .	100%
3	Provides evidence, explanation, and decisions, with coherent logic linking all three.	80%
2	Presents results or figures, but the explanation is weak or disconnected from the evidence.	50%
1	Only presents results without explanation, or the methodology is clearly flawed.	20%
0	No answer provided / Blank.	0%

Appendix

- Demo will take place on **Tuesday, November 10**, together with the HW2 demo.
- Demo details will be announced later.
- For homework-related issues, please **post on the eeclass discussion forum**, and the TAs will reply ASAP.
- For personal matters or private inquiries, you can reach the TA at george930502@gapp.nthu.edu.tw.

Q & A